



The Collaborative Ocular Melanoma Study Randomized Trial of Iodine 125 Brachytherapy for Choroidal Melanoma

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At the time of the Collaborative Ocular Melanoma Study (COMS) (see page S148), enucleation was the standard of care for uveal melanoma. However, questions existed about the safety of this treatment option and whether it accelerated metastatic death by disseminating tumor cells into general circulation.¹ In the late 1970s, plaque brachytherapy emerged as a conservative alternative for uveal melanoma treatment, intensifying the debate. Some clinicians raised concerns that irradiated eyes still showed viable melanoma cells on histologic examination.² However, the overarching question quickly became: Does enucleation or brachytherapy provide a safer and more effective option for treatment of uveal melanoma patients? This debate contributed to the launch of the COMS. Today, the COMS remains the largest study ever performed in ocular oncology, with 43 participating centers, more than 2000 recruited patients, and 28 publications.

The COMS was divided into 3 trials or observational studies based on the size of the choroidal melanoma; the trial on medium-sized melanomas focused on the enucleation-versus-brachytherapy debate. This trial randomized 1317 patients to enucleation or iodine 125 (I^{125}) brachytherapy, ultimately finding no significant differences in survival rates between the groups through 12 years of follow-up.³ This study proved that mortality and quality of life were the same after enucleation and brachytherapy, providing reassurance to both ophthalmologists and patients.

As brachytherapy became a more commonly accepted treatment, a need emerged for deeper investigation of its efficacy and morbidity. Again, the COMS provided a unique opportunity to evaluate this question in a large, well-characterized group of patients with an extremely low misdiagnosis rate (5/1532 cases). Researchers examined the 650 patients in the brachytherapy group for frequency and predictors of subsequent enucleation and treatment failure.⁴ They showed that enucleation and local treatment failure were relatively uncommon events after brachytherapy, with almost 90% of patients exhibiting tumor control and 88% retaining the eye at 5 years. These findings, in combination with the prior study showing similar survival outcomes in enucleation and brachytherapy groups, provided further support for brachytherapy as a safe, effective treatment option that can conserve the globe. Today, brachytherapy is accepted commonly as first-line treatment for medium-sized choroidal melanoma. These results also set the stage for subsequent use of radiation for treatment of a variety of ocular diseases, using a range of

therapies such as ruthenium 106, proton beam, and stereotactic radiotherapy.

Although this study resolved the enucleation-versus-brachytherapy debate, questions linger today. Some researchers question if the follow-up times in this study (rarely more than 10 years) were long enough to detect the development of all metastases. It is also possible that ocular morbidity after brachytherapy was caused by specific use of I^{125} . In addition, the original categorization of patients for the COMS was based solely on tumor size, although today, measures of uveal melanoma prognosis are known to be multifactorial.⁵ The presence of subretinal fluid, orange pigment, absence of drusen, and low reflectivity on A-scan ultrasound are important prognostic indicators of uveal melanoma versus less suspicious pigmented lesions that can be observed. Thus, the medium-sized melanoma group in the COMS likely contains a heterogeneous range of tumors, making it difficult to determine exactly how much the choice of treatment method affected patient outcomes compared with the genetic profile of the melanoma.

Nevertheless, we still need local treatment of medium (and large) melanomas to target the primary tumor. The contribution of this article is to show that brachytherapy can meet this need safely and effectively. Looking forward, we must direct more resources toward combating the high mortality rate from metastases. In the COMS, the liver was the most frequently reported site of metastasis, although multiple sites were reported for many patients. The result is that the mortality rate of uveal melanoma remains largely unchanged from the mid-19th century. With cutaneous melanoma now being treated more effectively, uveal melanoma represents approximately 5% of melanoma diagnoses, but 20% of melanoma mortality.

We have reason to hope. New tools now exist that can risk-stratify patients based on gene expression profile, clinical profile, histopathologic characteristics, or a combination thereof. Liquid biopsy of blood also can provide a means to detect circulating tumor cells. Patients with micrometastatic disease can be considered for early intervention with systemic or organ-specific (liver) adjuvant chemotherapeutic, immunotherapy, or precision biologic treatment. Finally, discovery of new pathways and development of targeted biologics, such as mitogen-activated protein kinase (MEK) inhibitors, also can contribute to personalized medicine in these patients. The sum of all these approaches will bring us closer to our